Forecasting Selected European Power Price Spreads with Machine Learning and SHAP

Glen Kurokawa, August 2026

Abstract

We forecast two day-ahead electricity price spreads on the German borders —Germany minus France (DE-FR) and Germany minus Poland (DE-PL)— one day ahead, and ask whether modelling a spread directly beats differencing two separate country price forecasts. Using hourly ENTSO-E data from 2019 to 2026 and a walk-forward design with monthly retraining, we compare LASSO and gradient boosting (XGBoost) and test differences with the Diebold–Mariano test. On DE-FR the two approaches tie. On DE-PL, direct modelling with gradient boosting is best at 15.39 €/MWh mean absolute error, a statistically significant improvement over both differencing and LASSO. SHAP attributions and raw-data checks trace the edge to renewable-driven threshold effects —negative German prices under high wind and solar, and Poland flipping from importer to exporter— that a spread model can represent but two separate price models cannot. The model runs as a live, self-updating forecasting service.

Keywords: electricity price spreads • day-ahead forecasting • gradient boosting • SHAP • Diebold–Mariano test • cross-border congestion

1. Introduction

A *price spread* is the difference between two neighbouring countries' electricity prices in the same hour. European power markets are linked by cross-border interconnector cables. When a cable has spare capacity, trade tends to bring the two countries' prices close together; when the cable is full ("congested") the prices can split apart and a spread opens. This project forecasts two such spreads one day ahead: Germany minus France (DE-FR) and Germany minus Poland (DE-PL).

There are two ways to forecast a spread. One can model it directly, training a single model on the price difference. Or one can difference two forecasts, forecasting each country's price on its own and then subtracting. Both aim at the same number. The question this study answers is which works better for the spreads examined here, and whether the answer depends on the border.

The answer differs by border. For DE-FR the two methods tie: every approach lands around €19/MWh of average error. For DE-PL, modelling the spread directly with gradient boosting is best, at €15.39/MWh average error, a statistically significant improvement over both differencing and LASSO regression. SHAP (introduced in Section 2) shows why. The DE-PL spread has sharp switch points, for example the hours when abundant German wind and solar output pushes German prices below zero while Polish coal plants hold a price floor, so the spread jumps rather than sliding smoothly. A straight line cannot represent a jump like that, but decision trees can. Those switch points occur on the DE-PL border, which is more congested, and not on the well-connected DE-FR one, which is why the extra modelling effort pays off there.

The model also runs as a live system: a daily job fetches new market data, retrains, and updates a public dashboard (Section 8). The rest of the document covers the background (Section 2), the data (Section 3), the method (Section 4), results for each model (Sections 5 and 6), the SHAP interpretation (Section 7), the live system (Section 8), and conclusions (Section 9).

2. Literature Review

Most electricity-price forecasting research targets a single country's price, not a spread. A widely cited review (Lago et al., 2021) provides two reference models: LEAR, a LASSO regression, and a deep neural network. In their benchmark the neural network is generally more accurate, but LEAR is fast and competitive, which makes a LASSO baseline demanding. This study uses that same LASSO approach as its benchmark.

Day-ahead price data is tabular and medium-sized (about 66,000 hourly rows here). On tabular data of this size, tree-based models remain state-of-the-art and generally outperform deep neural networks, whose advantage shows on unstructured data such as images and text (Grinsztajn et al., 2022). This does not contradict Lago's result: their benchmark showed a neural network beating a LASSO baseline, not a tree-based one, which they did not test. The two findings compare different pairs, and the head-to-head that does exist favours trees here. The study therefore uses LASSO regression and gradient boosting, and leaves deep learning aside.

2.1. Explainability (SHAP), and why it is not a novel contribution

SHAP (Lundberg and Lee, 2017), which adapts the Shapley value from cooperative game theory to share a prediction fairly among the inputs, splits a prediction into a contribution from each input, showing what pushed the forecast up or down and by how much. It has been used to explain electricity prices: for example Trebbien et al. (2023) on the German market, and Pesenti and O'Sullivan (2026), who study thirty-nine European bidding zones and find that neighbouring zones account for about 61% of a price's explanation on average.

Explaining two countries' prices already contains what is needed to explain their difference, so pointing SHAP at a spread is not a new idea. What the existing explainability work does not do is *forecast* future prices out-of-sample; those models use same-hour information that would not be known in advance. Here, SHAP is applied to a forecasting model only after it has been shown to work, both to interpret what drives the forecasts and, in Section 7, to investigate why modelling the spread directly outperforms differencing, though that analysis describes the structure the model learned rather than establishing causation in the market.

2.2. Why a spread can behave differently from a price

When the cross-border cable has spare capacity, trade tends to pull the two countries' prices together and the spread stays small; it can also sit near zero simply because both prices move up or down together, rather than because the cable is doing the work. When the cable is congested, the prices diverge and the spread widens. So a spread is usually small and opens up mainly during congestion. Alasseur and Féron (2018) give a formal model of this. This is the reason to expect direct modelling to win: forecasting each country's price separately spends most of the model's effort on the big shared drivers (fuel prices, weather, total demand) that cancel out when the two prices are subtracted, whereas a model trained on the spread itself can concentrate on the congestion behaviour. Whether that advantage shows up is the empirical question here.

The closest existing work describes prices rather than forecasts them. Saez et al. (2019) study when neighbouring prices converge but do not forecast the spread. Earlier forecasts of a spread are limited: Ferkingstad and Løland (2014) forecast a Nordic contract with classical statistics, and Imani (2024) predicts only the *direction* of Italian inter-zonal congestion (the sign of the price difference), not its size.

2.3. What this study adds

The specific question left open is: for a major European spread, does modelling it directly forecast it better than differencing two separate country forecasts? Both predict the same number. A "no" answer would be informative too, showing the spread holds no extra structure beyond the two prices. No prior study was found that forecasts a European spread out-of-sample and interprets it with SHAP. This study fills that gap: it forecasts two major German-border spreads out-of-sample, tests directly whether modelling the spread itself beats differencing two country forecasts, explains the outcome with SHAP, and runs the result as a live system that retrains each day.

3. Data

The data comes from the ENTSO-E Transparency Platform, the public source for European electricity data, pulled through its API with the `entsoe-py` Python client. For each of the three countries, Germany-Luxembourg (DE-LU), France (FR) and Poland (PL), it covers the day-ahead price, actual demand and the day-ahead demand forecast, actual generation by fuel type, the day-ahead wind and solar forecast, the day-ahead total generation forecast, installed capacity, and power plant outages. Cross-border data covers physical flows on all of Germany's borders, the day-ahead scheduled trades across them, and France's and Poland's net positions (exports minus imports). The two forecast targets are the spreads: spread_DE_FR = price_DE - price_FR and spread_DE_PL = price_DE - price_PL.

Inputs follow a simple naming pattern used in the charts. For example `wind_solar_fc_PL` is Poland's day-ahead wind and solar forecast, `load_fc_DE` Germany's demand forecast, `gen_fc_FR` France's generation forecast, `outage_DE` German plant outages, `net_pos_PL` Poland's net position, and `sched_exch_DE_PL` the scheduled Germany-to-Poland exchange.

Turning this data into one clean table meant fixing four data problems. First, the German bidding zone in its current form only started in October 2018, so neither spread can begin before then; the assembled table starts on 1 January 2019. Second, some Polish prices came through in złoty rather than euros; these were converted using monthly European Central Bank exchange rates. Third, from 1 October 2025 several markets switched to 15-minute pricing, so prices now arrive four times per hour while everything else is hourly; all series were averaged onto a common hourly grid. Fourth, formatting issues (mixed daylight-saving time stamps, awkward column headers, and outages recorded as start and end events) were cleaned up, with outages converted into an hourly "megawatts offline" figure. The result is one hourly table of about 66,000 rows and 51 columns, from January 2019 onward.

4. Methodology

The day-ahead market runs one auction per day, setting prices for all 24 hours of the next day. Bidding closes at noon, and the cleared prices, cross-border schedules and net positions are published shortly after. The forecast is made before then, predicting tomorrow's 24 hourly spreads from what is known at that point. One model is trained for all hours rather than one per hour, with hour of day, day of week and month supplied as inputs so it can still learn time-of-day patterns.

The model's inputs are the day-ahead forecasts of demand, wind and solar, and generation for the three countries, plant outages, calendar and public holiday flags, and the spread's own value 24, 48 and 168 hours earlier (its "lags", all at least a day old and so known when the forecast is made). The scheduled cross-border exchanges and net positions are left out: they are set by the auction itself, so they are not available when the forecast is made. Section 6 adds them as a quasi-perfect-foresight benchmark to measure what they would be worth.

The benchmark model is LASSO. Accuracy is measured mainly by *mean absolute error* (MAE), the average size of the forecast miss in €/MWh, and by root-mean-square error (RMSE), which penalises large misses more. These are compared against three naive yardsticks: predict zero, predict the historical average, and "persistence" (tomorrow equals the same hour yesterday). A Sharpe ratio is not reported: as Section 5 explains, under the assumptions here it would take values too large to be informative.

Two ways of splitting the data were tried. A one-off split (train to 2023, test 2025 onward) failed, because the market changed character partway through (Section 5). It was replaced by walk-forward evaluation: the model is retrained each month on a rolling two-year window and used to forecast the following month. Development used 2021 to 2024; the final test period (2025 to mid-2026) was set aside and scored once.

4.1. Gradient boosting, tuning, and significance testing

The second model is gradient boosting, implemented with XGBoost. Its inputs are not rescaled or one-hot encoded, so the calendar variables are passed directly. Everything else matches the LASSO setup (same walk-forward, same inputs, same test rows) so any difference in results comes from the model choice alone. Within each monthly retrain, the last eight weeks of the training window are held back to decide when to stop adding trees.

The settings were chosen in two steps: a manual sweep to find the rough range, then an Optuna search minimising the walk-forward error. The search validated on rolling recent windows (2023 H2, 2024 H1 and 2024 H2 in turn) so the tuning was not dominated by the 2022 energy crisis. The whole search was logged with MLflow. These settings were tuned once and reused for both the forecast and the quasi-perfect-foresight model; the flat accuracy floor in Section 6 makes that safe, since re-tuning for a smaller input set moves the error by less than it risks over-fitting the choice to the test period.

To check whether one model is more accurate than another rather than luckier on this test period, results are compared with the Diebold-Mariano test. Because a once-a-day forecast produces errors correlated over several days, the test uses a Newey-West variance estimate that accounts for that correlation. This estimate depends on a bandwidth: the number of lags of error correlation it allows for. Two standard choices anchor what is reasonable here. The Diebold-Mariano convention sets the bandwidth to one less than the forecast horizon, which for a one-day-ahead forecast is essentially a single day, enough to absorb the 24 correlated hourly errors that come from one auction. The automatic Newey-West rule, about $4(T/100)^{2/9}$ lags, works out to roughly a week on this test sample. The result is therefore reported across a range of bandwidths, from one day up to a conservative two-week window: two weeks is two to three times the automatic choice, and because a longer bandwidth widens the standard error and makes significance harder to reach, a difference that stays significant out to two weeks is not an artefact of too short a window. The test is applied both to direct versus differenced within each model, and to XGBoost versus LASSO on the directly modelled spread.

5. Results: LASSO Regression

The market changed character over the 2018 to 2026 sample. The DE-FR spread was near zero or negative from 2018 to 2023 (Germany usually the cheaper, positive in only 7 to 28% of hours), then turned positive in 2024 to 2026 (typically +18 to +21 €/MWh, positive about 70% of the time) as French nuclear output recovered while German prices stayed tied to expensive gas. Wind and solar inputs also trended upward as capacity was built.

Because of that shift, training once on old data failed. A LASSO trained on 2018 to 2023 mis-predicted the new regime and was beaten by the naive yardsticks on the 2025+ test set: a DE-FR error of 42.8 €/MWh against a persistence yardstick of 24.4 €/MWh, and a DE-PL error of 23.1 €/MWh, worse than persistence (22.0 €/MWh) and worse than predicting zero. Retraining each month on a rolling two-year window fixed this, and both models then beat every yardstick:

Retraining each month improved the metrics, and adding lags further improved them.

Now the central question: is modelling the spread directly better than forecasting the two prices separately and subtracting? Using the same walk-forward LASSO with lags on the 2025+ test rows:

Both gaps are around €1/MWh or less on errors of €16–19/MWh, so a Diebold-Mariano test checks whether either difference is real or noise. The verdict differs by border:

- DE-FR: no real difference. The direct model is nominally worse by about €0.11/MWh, and no sensible setting makes that significant; forecasting the two prices separately and subtracting loses nothing here.

- DE-PL: direct is better, by €0.71/MWh, and this holds under every setting tested, including a conservative two-week one ($p = 0.000$ to 0.005).

So the advantage of direct modelling manifests in DE-PL and not DE-FR, albeit only modestly. The likely reason is that the Polish price is harder to forecast on its own, so differencing two separate forecasts stacks up two lots of error, whereas modelling the spread directly cancels the shared drivers and focuses on the congestion behaviour, the mechanism from Alasseur and Féron (2018) that motivated the study.

A Sharpe ratio is not reported. Had one been computed under the assumptions here (take a position whenever the predicted spread exceeds €5/MWh, charge a €1/MWh round-trip cost, sum the hourly profit into daily figures and annualise), it would have run to roughly 15 to 24. Numbers that size are not a real trading edge. Two things inflate them

Table 1. Walk-forward LASSO mean absolute error against naive benchmarks, 2025+ test set.

Approach	DE-FR MAE (€/MWh)	DE-PL MAE (€/MWh)
persistence (t-24h)	24.4	22.0
static LASSO	42.8	23.1
walk-forward, no lags	22.4	17.5
walk-forward, with lags	19.4	16.4

Table 2. Direct versus differenced walk-forward LASSO (with lags), mean absolute error.

Spread	Direct MAE (€/MWh)	Differenced MAE (€/MWh)
DE-FR	19.4	19.3
DE-PL	16.4	17.1

at once: they are a signed accumulation of spread levels in €/MWh rather than a risk-normalised return, so their mean-to-standard-deviation ratio is not a financial Sharpe; and the spread stayed almost entirely one-signed through the test period, making its direction trivially easy to call and the strategy profitable on nearly every day. A Sharpe above about 2 is already exceptional, so figures in the twenties are uninformative, which is why MAE is the metric to trust. Whether gradient boosting widens the DE-PL advantage is the subject of Section 6.

6. Results: Gradient Boosting

Tuning gradient boosting hit an accuracy floor, not a capacity ceiling. In the manual sweep, making the trees deeper (depth 2 to 8) left the validation error stuck at about €18/MWh even as the training error fell from 24 to 11 €/MWh, the classic sign of a model memorising noise rather than forecasting better. The Optuna search confirmed it: across sixty attempts the validation error never left a band of about €17.0 to €18.0. The best setting was a deep, heavily restrained tree, yet it reached the same floor as a shallow, unrestrained one. Two opposite configurations landing on the same error, all sixty within about €1/MWh of each other, means there is a real floor to how well this data can be forecast. This matches the Section 2 point that, for data of this size and shape, added complexity buys little.

Run once on the untouched 2025+ test set, tuned XGBoost against the LASSO benchmark, for both spreads and both methods:

Two significance tests make the comparison precise. Direct versus differenced within XGBoost: on DE-PL, direct beats differencing by €0.87/MWh, significant for every error-correlation window up to a conservative two weeks ($p = 0.000$ to 0.020); on DE-FR direct also "wins" nominally, but only for short windows, losing significance once the test allows for a week or more of error correlation, and the win comes from differencing two wobbly gradient-boosting forecasts stacking up their errors (XGBoost-differenced is the worst DE-FR row). XGBoost versus LASSO on the direct spread: on DE-FR the two are tied (significant at no window), but on DE-PL gradient boosting beats LASSO by €1.00/MWh, significant at every window ($p = 0.000$ to 0.004).

The gains from modelling the spread directly and from using gradient boosting appear only on DE-PL, not DE-FR. The improvement is in typical-hour accuracy, not on every measure: the direct XGBoost's RMSE (26.76 €/MWh) is marginally higher than the direct LASSO's (25.82 €/MWh), so it makes smaller misses on most hours but a few larger ones on the rare extreme congested hours. MAE is the more relevant measure when the typical hour's miss matters more than the occasional large one. The RMSE difference is small and, on a Diebold-Mariano test of squared-error loss, not significant for any window from one day to two weeks ($p = 0.16$ to 0.39). The squared-error series is dominated by a handful of extreme congested hours, so a gap this size is within noise. The direct XGBoost is therefore better on MAE and not measurably worse on RMSE.

These are two separate tests of two separate nulls, not one test applied twice. The Diebold-Mariano test compares expected loss under a chosen loss function, so the absolute-loss test asks whether the models have equal MAE and the squared-loss test whether they have equal MSE, the square of RMSE; they are reported separately because the two can disagree, as they do here. The RMSE result is a failure to reject, which a stricter multiple-testing threshold could only reinforce.

How much does forecasting before the auction cost? The excluded auction outputs, the scheduled cross-border exchanges and net positions, can be added to build a quasi-perfect-foresight benchmark. It is not a forecast, since it needs data published at the same moment as the price; and it is only quasi-perfect, because its demand, wind, solar and generation inputs are still day-ahead forecasts rather than realised values, so it has quasi-perfect rather than perfect information. With those inputs included, the direct DE-PL XGBoost reaches 14.40 €/MWh, about €1/MWh better than 15.39 €/MWh. So forecasting before the auction costs roughly one euro per megawatt-hour of accuracy. Quasi-perfect foresight also widens the direct model's edge over differencing, from €0.87/MWh to €2.23/MWh, which Section 7 traces to the specific cross-border thresholds those inputs

Table 3. Out-of-sample accuracy on the untouched 2025+ test set: LASSO and XGBoost, direct and differenced.

Spread	Model / method	MAE (€/MWh)	RMSE (€/MWh)
DE-FR	LASSO direct	19.42	26.83
	LASSO differenced	19.30	27.16
	XGBoost direct	19.28	27.20
	XGBoost differenced	19.98	27.92
DE-PL	LASSO direct	16.39	25.82
	LASSO differenced	17.10	26.03
	XGBoost direct	**15.39**	**26.76**
	XGBoost differenced	16.26	25.72

expose.

To put these errors in scale: over the test period the DE-PL spread has a standard deviation of 32.5 €/MWh and lies within ± 1 €/MWh of zero about 22% of the time, so it swings widely and often changes sign. Against that, the 15.39 €/MWh MAE is about 47% of one standard deviation. Using the RMSE, the model explains roughly 32% of the spread's variance. Percentage errors such as MAPE are not reported, because the spread crosses zero and dividing by near-zero values makes them unstable (Lago et al., 2021).

This is the model interpreted with SHAP in Section 7.

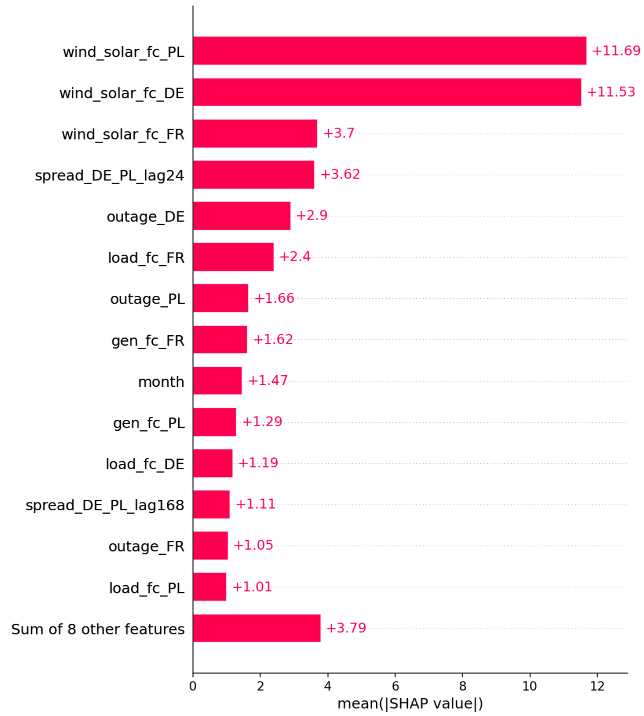

Figure 1. How much each input moves the forecast on average (mean absolute SHAP, €/MWh). The two renewable forecasts dominate; the auction outputs are absent by design.

7. Interpretation: What the DE-PL Model Learned

SHAP was applied to the winning model, the direct tuned XGBoost for DE-PL. A single explanatory model was trained on 2023 to 2024 and its predictions on 2025+ broken down with SHAP.

With the auction outputs removed, the forecast is driven almost entirely by the renewable forecasts. Polish wind and solar (worth on average 11.7 €/MWh of forecast movement) and German wind and solar (11.5 €/MWh) tower over everything else. A long way behind come French wind and solar (3.7 €/MWh), the spread's own value from yesterday (3.6 €/MWh), German and Polish plant outages (2.9 and 1.7 €/MWh), French demand (2.4 €/MWh), the generation forecasts, and the month of the year (1.5 €/MWh). Calendar inputs otherwise barely matter, which fits a spread driven by physical fundamentals rather than the clock. The directions make economic sense: high Polish wind and sun makes Polish power cheap and widens the (Germany minus Poland) spread, while high German wind and sun makes German power cheap and narrows it. These signs agree with LASSO's coefficients, so the two models tell the same story. Renewables dominate more here than under quasi-perfect foresight (Section 7.2), because without the scheduled flows the forecast reads the congestion signal off the fundamentals that drive it.

The spread's own recent values sit mid-table (the 24-hour lag at 3.6 €/MWh, the week-old lag near the bottom at 1.1 €/MWh), lower than their usefulness for accuracy would suggest. Two things explain the gap. First, the SHAP ranking measures *amplitude*: how many €/MWh an input typically moves the forecast. A lag acts as a small, usually-correct persistence nudge that trims many hours by a little without ever swinging the forecast far, so it lowers the average error while barely registering on an amplitude scale, whereas a renewable driver occasionally moves the forecast by tens of €/MWh and dominates the ranking. Second, the spread is only weakly autocorrelated: its correlation with its own value a day earlier is only about 0.3, and weaker still a week back, so each lag carries limited information, most of it in the 24-hour lag. The lags are worth keeping but form a diffuse, low-amplitude signal, which is exactly the profile that looks minor in a SHAP bar chart while still helping accuracy. Measuring that accuracy contribution needs an ablation rather than SHAP, the test described in Section 7.2.

This also explains why gradient boosting beats LASSO.

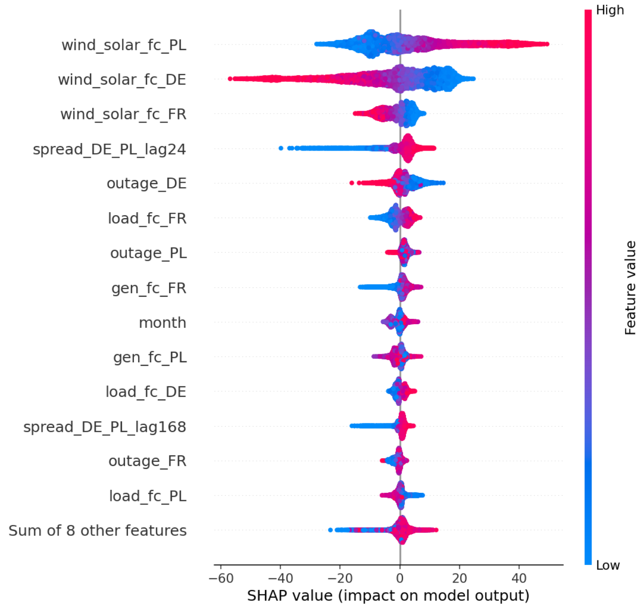

Figure 2. Each dot is one hour. Left-right shows how much that input pushed the forecast that hour; colour shows whether the input was high (red) or low (blue). High Polish renewables push the spread up; high German renewables push it down.

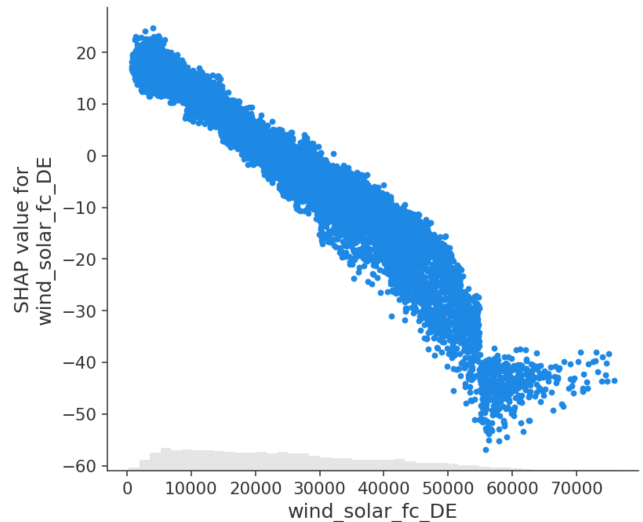

Figure 3. German wind and solar forecast: a smooth decline that steepens into a cliff near 55 GW, where German prices turn negative.

The renewables act through sharp thresholds, and LASSO, fitting only straight lines, cannot bend at a threshold. The clearest is German wind and solar: as its forecast climbs past about 55 GW, German prices fall through zero and the spread drops steeply rather than sliding (Section 7.1). Polish renewables behave similarly where a surplus tips Poland from importing to exporting and its price collapses. These are switches, not gradients, and they are the congestion behaviour Alasseur and Féron (2018) describe. This is the mechanism behind the Section 6 result, present on DE-PL and absent on DE-FR.

7.1. Two thresholds, checked against the raw data

Two of these patterns are worth checking directly against the raw data, not just trusting the model. In both cases the data confirms the pattern is a real economic effect — one visible directly in the raw market data and explained by how the market physically clears, not something the model fit to noise — and both show why a spread has to be understood as one joint object, not two separate prices.

German wind and solar: a negative-price cliff near 55 GW. The model shows the effect declining smoothly, then dropping steeply around 55 GW of forecast wind and solar. Grouping German prices by that forecast shows why:

Below the threshold, each extra gigawatt of renewables pushes out a fossil plant and trims a still-positive price. Around 55 GW the fossil plants are all gone, and any more supply pushes the German price *below zero* (a real feature of power markets, where some plants must keep running and are paid to keep producing). Negative-price hours jump from about a third to a majority. Poland, mostly coal and with a price floor, does not follow, so the spread blows out downward. This is a threshold a straight line cannot draw, and it is the model's single largest driver.

Polish renewable surplus: the spread is widest when Poland has little to export, then jumps as Poland becomes an exporter. Whether Poland imports or exports — its net position — captures this most directly, but the net position is an auction output, known only when the price is, so the model cannot use it as an input; instead it reads the same effect through the Polish wind and solar forecast, which drives the surplus that turns Poland into an exporter. The effect can still be checked in the raw data by grouping the actual DE-PL spread by net position over 2023 to 2024:

The spread is most negative when trade is balanced and gets *less* negative as Poland imports more heavily, the opposite of the naive guess that an importing, expensive Poland should show the smallest gap. The reason is that the spread is a difference. Heavy Polish importing happens in the cold, still, high-demand hours when renewables are low everywhere, so Germany is also running expensive plants; both prices are high and close together, and the gap shrinks. In balanced conditions Germany runs its cheap renewables while Poland runs coal, so the gap is widest. The step comes when a Polish renewable surplus tips the country into exporting and its own price collapses, which the model captures from the Polish renewable forecast that drives the surplus, and which gradient boosting can bend at the threshold while a straight line cannot.

In both cases the usable structure is a threshold rather than a smooth slope, which gradient boosting can fit and LASSO cannot; this is what SHAP shows directly, and it explains why XGBoost beats LASSO on this border. Both thresholds are also joint, spanning the two zones at once (German supply

Table 4. Actual German day-ahead prices grouped by wind-and-solar forecast, 2023–2024.

German wind+solar forecast (GW)	mean price_DE (€/MWh)	share of hours price_DE < 0 (%)	mean spread (€/MWh)
20–35	79	2	-20
45–50	30	23	-36
50–55	14	34	-48
55–60	**-7**	**59**	**-60**
60+	-28	89	-65

Table 5. Actual DE-PL spread grouped by Poland's net position, 2023–2024.

Polish net position (MW)	mean actual spread (€/MWh)
heavy import (-6000. . . -2000)	-19.5
moderate import (-1000. . . -500)	-22.8
balanced (-1. . . +1)	**-25.5**
light export (+1. . . +500)	-9.8
heavy export (+2000. . . +6000)	+10.7

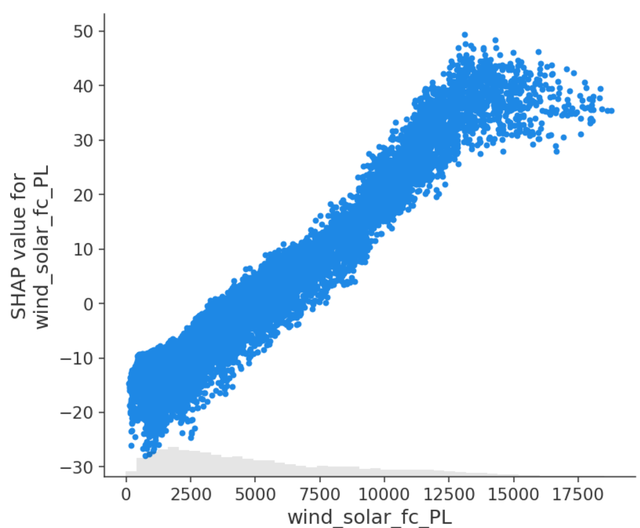

Figure 4. Polish wind and solar forecast: the spread rises as Polish renewables climb, steepening as a surplus turns Poland into an exporter and collapses its price.

against a Polish price floor, and Polish supply against German prices), which is why a model trained on the spread can represent them, and part of why direct modelling beats differencing even without the auction outputs.

7.2. Why direct modelling wins, and what the auction outputs add

Section 6 showed the direct spread model beating differencing on DE-PL by €0.87/MWh, and by €2.23 with quasi-perfect foresight. SHAP explains both, because the differenced forecast is price_DE_forecast minus price_PL_forecast and SHAP contributions add up: differencing's implied attribution of any input to the spread is, hour by hour, the German price model's SHAP value minus the Polish one's. So for each input one can compare how the direct model attributes it to the spread against how differencing implicitly does.

The two dominant drivers are single zone. Polish and German wind and solar each cheapen their own country's power, so the separate price models learn them independently and differencing reproduces them almost exactly, the direct and differenced attribution curves matching in shape. The direct model gains nothing on the renewables. Its €0.87 MAE edge is therefore mostly the general penalty of differencing noted in Section 5: the spread built from two separate price forecasts carries both models' errors.

Adding the two auction outputs — net_pos_PL (Poland's net position, its exports minus imports) and sched_exch_DE_PL (the power scheduled to flow from Germany to Poland) — as inputs to both the direct spread model and the two price models used for differencing shows where the rest of the direct model's advantage comes from. Both variables describe the relationship between the two countries rather than either one alone, so they are hard to derive from German or Polish data on its own. This raises the direct model's edge over differencing from €0.87/MWh to €2.23/MWh. Figure 5 shows why for net_pos_PL: the direct model's attribution to the spread jumps by 8.5 €/MWh where Poland crosses from importer to exporter, while the differenced line moves only 2.6 €/MWh. Figure 6 shows the same for sched_exch_DE_PL: the direct model captures a 6.3 €/MWh drop at the roughly 500 MW congestion point, against 0.5 €/MWh for differencing.

The figures show how the trained model attributes these inputs, but attribution alone is not proof that they drive accuracy: an input can be assigned a large SHAP value without actually improving the forecast. An ablation tests that directly. Rather than explaining a fixed model, it deletes the two cross-border inputs, retrains both the direct and differenced approaches from scratch so each reorganises around what remains, and measures the change in out-of-sample error. This shows the two inputs account for about 45% of the direct model's advantage over differencing; the rest is the general differencing penalty, which applies whatever the inputs and is the part the forecast keeps.

Two caveats. SHAP describes what the model *learned*, not proven cause and effect, and these explanations come from representative explanatory models rather than any single day's live forecast.

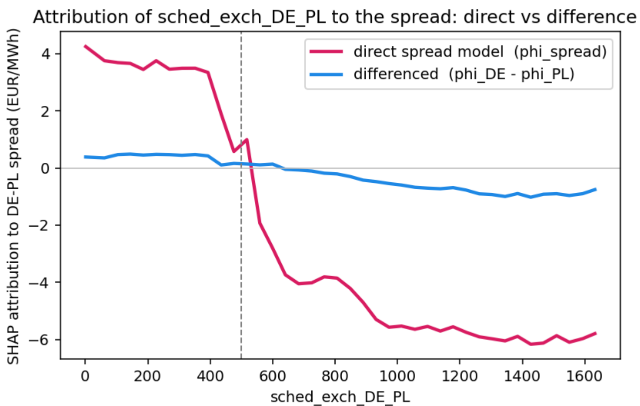

Figure 5. Quasi-perfect foresight. How the Polish net position is attributed to the DE-PL spread. The direct model (pink) steps up sharply as Poland turns exporter; the differenced approach (blue), the German price model's SHAP minus the Polish one's, barely moves.

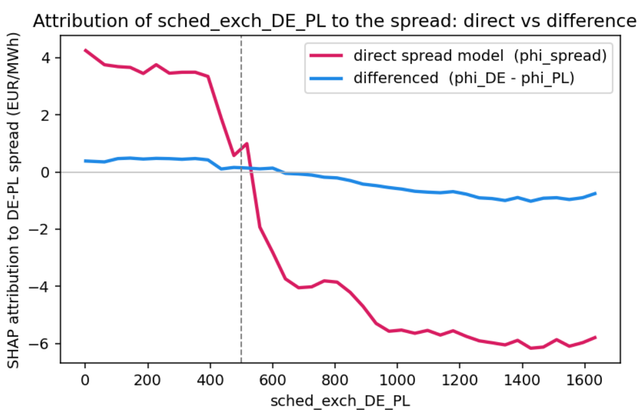

Figure 6. Quasi-perfect foresight. The scheduled Germany-to-Poland flow. The direct model captures the congestion drop near 500 MW; differencing captures a fraction of it.

8. Deployment: A Live Forecasting System

The model runs as a scheduled daily job: it fetches new market data, retrains, and writes the next day's forecast to a cloud database that a public web dashboard reads. The live pipeline uses the same research code; it just works on the most recent days.

Each daily run checks the latest stored date, re-fetches a short recent window from ENTSO-E (the last ten days, so ENTSO-E's frequent late corrections to demand, generation and outages get picked up rather than frozen at their first value), rebuilds those hours into the research table format, and writes them to a cloud database. The run then retrains on the last two years and writes the coming day's forecast. Because the model uses only inputs published before gate closure, the run happens in the late morning, before the auction, and forecasts the next day's 24 spreads before the market sets them.

9. Conclusion

The study set out to answer one question: for the day-ahead spreads on these two German borders, is a spread forecast better by modelling it directly or by differencing two separate country forecasts? The answer depends on the border. On DE-FR the two methods tie and nothing beats LASSO. On the DE-PL price spread, modelling the spread directly with tuned gradient boosting is best, at 15.39 €/MWh average error, beating both the differenced forecast and LASSO with statistical significance. Allowing quasi-perfect foresight improves accuracy by only about €1/MWh.

The gains from direct modelling and from gradient boosting appear *together* and *only* on DE-PL, and SHAP traces them to renewable-driven switch points: German wind and solar pushing prices below zero, and a Polish renewable surplus flipping the country from importer to exporter.

It covers two spreads on two German borders over one test period; no Sharpe ratio is reported; and SHAP shows what the model learned, not causation. Natural next steps are more borders and testing under realistic trading costs. The project is a tested, continuously-running forecasting system.

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